Market yield expectations and forward yields

Understanding the Yield Curve (Chapter 2)

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INTRODUCTION

> introduction

Our recent report *Overview of Forward Rate Analysis* introduced a series on the theme *Understanding the Yield Curve* . It argued that three main forces determine the term structure of forward rates: the market's rate expectations; required bond risk premia; and the convexity bias. Separate reports discuss each of these forces. This report focuses on the impact of the market's rate expectations on the yield curve shape.

> Our recent report, "An Overview of Forward Yield Analysis," introduced the theme of our "Understanding the Yield Curve" series. Three main factors determine the term structure of forward yields: market yield expectations, bond risk premiums, and convexity bias, which will be discussed in separate reports. This report focuses on the impact of market yield expectations on the shape of the yield curve.

The impact of rate expectations on today's yield curve shape is best isolated by assuming that the pure expectations hypothesis holds. According to this hypothesis, all government bonds have the same near-term expected return (that is, all bond risk premia are zero). If the near-term expected returns are equal across maturities, initial yield differences must offset any expected capital gains or losses that are caused by the market's rate expectations. For example, if the market expects rates to rise and the long-term bonds to suffer capital losses, these bonds must have an initial yield advantage over the one-period bond (to offset the expected capital losses). Therefore, expectations of rising rates tend to make today's yield curve upward sloping. Conversely, expectations of declining future rates tend to make today's yield curve inverted. In a similar way, the market's expectations of future curve flattening or steepening influence the curvature of today's yield curve.

> To explain the impact of yield expectations on the current shape of the yield curve, it is best to assume the perfect expectations hypothesis holds. According to this hypothesis, all government bonds have the same near-term expected return (i.e., all bonds have a zero risk premium). If the near-term expected returns of bonds with different maturities are equal, then initial yield differences must offset any expected capital gains or losses caused by market yield expectations. For example, if the market expects yields to rise and long-term bonds to suffer capital losses, these bonds must have an initial yield advantage over one-year bonds (to offset the expected capital losses). Therefore, expectations of rising yields tend to make the current yield curve upward sloping. Conversely, expectations of future yield declines tend to make the current yield curve inverted. Similarly, market expectations of a flattening or steepening curve in the future affect the curvature of the current yield curve.

We emphasize the distinction between the statements “the forwards imply rising spot rates” and “the market expects rising spot rates”. The first statement is related to the forward rates' role as break-even levels of future spot rates. By construction, the spot rate changes that the forwards imply for the next period are such rate changes that would make all government bonds earn the same one-period return. Whenever the spot rate curve is upward sloping, the forwards imply rising rates. That is, rising rates are needed to offset long-term bonds' yield advantage. However, it does not necessarily follow that the market expects rising rates. An upward-sloping spot rate curve also may reflect higher near-term required returns for long-term bonds than for the riskless one-period bond (so-called positive bond risk premia). The changes in spot rates that the forwards imply would be approximately equal to the expected spot rate changes only if the restrictive pure expectations hypothesis were true.

> We emphasize the distinction between "the implied rise in spot yield by forward yields" and "the market's expected rise in spot yields." The first statement relates to the forward yield as the break-even point for future spot yields. The change in spot yield implied by forward yields

> will ensure that all government bonds earn the same return during the same period. When the spot yield curve slopes upward, forward yields imply a rise in yields. That is, the rise in yields offsets the yield advantage of long-term bonds. However, this does not necessarily mean that the market expects a rise in yields. An upward-sloping spot yield curve may also reflect a higher short-term return for near-term long-term bonds than for risk-free one-year bonds (the so-called positive bond risk premium). Only when the restricted perfect expectations hypothesis is true will the change in spot yield implied by forward yields approximately equal the change in expected spot yields.

In this report, we also present empirical evidence about rate expectations and conclude that the pure expectations hypothesis is, in many ways, at odds with historical experience. We show that forward rates are poor predictors of future spot rates. In fact, long-term rates tend to move away from the direction implied by the forward rates. We also contrast the expectations "implied" in the forward rate structure to the expectations revealed in explicit surveys among bond market participants. The comparison suggests that forward rates are upward-biased measures of the market's rate expectations because the market appears to require higher expected returns for long-term bonds than for short-term bonds.

> In this report, we also present empirical evidence regarding yield expectations and conclude that the perfect expectations hypothesis is inconsistent with historical experience in many respects. We point out that forward yields do not accurately predict future spot yields. In fact, long-term yields often deviate from the direction implied by forward yields. We also compare the "implied" expectations in the forward yield structure with those disclosed in a survey of bond market participants. The comparison shows that forward yields overestimate market yield expectations, as the market seems to demand higher expected returns from long-term bonds than from short-term bonds.

ALGEBRAIC RELATIONS BETWEEN SPOT AND FORWARD RATES

> Algebraic relationship between spot yield and forward yield

This section reviews the relations between spot rates and forward rates and describes forward rates' role as break-even rates. The discussion in this section overlaps —— but expands on —— the discussion in *Overview of Forward Rate Analysis* . Readers familiar with the basic concepts in forward rate analysis may wish to move directly to the section "The Expectations Hypothesis and the Yield Curve."

> This section reviews the relationship between spot and forward yields and describes the role of forward yields as the break-even yield. The discussion in this section overlaps with, but expands upon, the discussion in "An Overview of Forward Yield Analysis." Readers familiar with the basic concepts in forward yield analysis may wish to skip directly to the "Expectations Hypothesis and Yield Curve" section.

Computation of Forward Rates

> Calculate forward yield

A spot rate is the discount rate of a single future cash flow such as a zero-coupon bond (zero). A coupon bond can be viewed and valued as a bundle of zeros. Given the price P_n of an n-year zero, the annualized n-year spot rate s_n can be computed as in Equation (1): [1]

> The spot yield is the discount rate for a single future cash flow, such as a zero-coupon bond. A coupon-paying bond can be viewed as the sum of a series of zero-coupon bonds. Given the price of an n-year zero-coupon bond... P_n The annualized n-year spot rate of return can be calculated as shown in equation (1). s_n:

$$P_n = \frac{100}{(1+s_n)^n}. \tag{1}$$

A forward rate is the interest rate for a loan between any two future dates, contracted today. The rate may be explicit in the price of a traded forward contract or it may be implicit in today's spot rate curve. A zero's discount rate (a multiyear spot rate) can be decomposed into a product of one-year forward rates. Thus, the spot rate is a geometric average of one-year forward rates: [2]

> Forward yield refers to the loan yield between any two future dates agreed upon now. The yield may be determined in the price of the forward contract or may be implied in the current spot yield curve. The discount rate of a zero-coupon bond (multi-year spot yield) can be decomposed into a product of a series of one-year forward yields. Therefore, the spot yield is the geometric mean of one-year forward yields:

$$(1+s_n)^n = (1+f_{0,1})(1+f_{1,2})(1+f_{2,3})\cdots(1+f_{n-1,n}), \tag{2}$$

where $f_{n-1,n}$ is the one-year forward rate between maturities n-1 and n, and $f_{0,1} = s_1$. If only one spot rate is known, all forward rates cannot be computed. However, if spot rates are known for each maturity, a given term structure of spot rates implies a specific term structure of forward rates. For example, if the m-year and n-year spot rates are known, $f_{m,n}$ (the annualized nm year rate m years forward) can be computed as in Equation (3):

> in $f_{n-1,n}$ It is the one-year forward yield between the terms n-1 and n. $f_{0,1} = s_1$ If only one spot yield is known, all forward yields cannot be calculated. However, if the spot yield for each maturity is known, the term structure of a given spot yield implies a specific term structure of forward yields. For example, if the spot yields for years m and n are known, they can be calculated as in equation (3). $f_{m,n}$ (Annualized forward yield over a period of m years):

$$(1 + f_{m,n})^{n-m} = \frac{(1 + s_n)^n}{(1 + s_m)^m}. \quad (3)$$

Figure 1 presents a hypothetical spot rate curve and two series derived using Equation (3): the implied forward path of the constant-maturity one-year rate at various future dates and the implied spot curve at one future date, one year hence. [3] It is important to distinguish the curve of one-year forward rates in column B from the implied spot rate curve one year forward in column C. Unfortunately the terminology varies because both are sometimes called the forward curves. To clarify the relations between these curves, Figure 2 shows the future years covered by ten spot rates (s_1 to s_{10}), ten one-year forward rates ($f_{0,1}$ to $f_{9,10}$) and nine implied spot rates one year forward ($f_{1,2}$ to $f_{1,10}$). The three curves are shown graphically in Figures 3 and 4.

> Figure 1 presents a hypothetical spot yield curve and two sequences derived using equation (3): the path of the one-year forward yield at different future dates, and the implied spot yield curve one year from now. It is important to distinguish the one-year forward yield curve in column B from the implied spot yield curve one year from now in column C. Unfortunately, both are sometimes referred to as forward yield curves. To illustrate the relationship between these curves, Figure 2 shows 10 spot yields over the next few years ($\ldots s_1$ arrive s_{10}), 10 one-year forward yields ($f_{0,1}$ arrive $f_{9,10}$) and 9 implied spot yields one year later ($f_{1,2}$ arrive $f_{1,10}$ These three curves are shown in Figures 3 and 4, respectively.

A		B		C		D = C - A	
Spot Rate Today(%)		One-Year Forward Rate(%)		Implied Spot Rate One Year Forward(%)		Implied Change in the Spot Rate(%)	
s_1	6.00	$f_{0,1}$	6.00	$f_{1,2}$	8.01	Δf_1	2.01
s_2	7.00	$f_{1,2}$	8.01	$f_{1,3}$	8.64	Δf_2	1.64
s_3	7.75	$f_{2,3}$	9.27	$f_{1,4}$	9.09	Δf_3	1.34
s_4	8.31	$f_{3,4}$	10.02	$f_{1,5}$	9.43	Δf_4	1.12
s_5	8.73	$f_{4,5}$	10.44	$f_{1,6}$	9.67	Δf_5	0.94
s_6	9.05	$f_{5,6}$	10.65	$f_{1,7}$	9.85	Δf_6	0.80
s_7	9.29	$f_{6,7}$	10.72	$f_{1,8}$	9.97	Δf_7	0.68
s_8	9.47	$f_{7,8}$	10.72	$f_{1,9}$	10.06	Δf_8	0.59
s_9	9.60	$f_{8,9}$	10.67	$f_{1,10}$	10.12	Δf_9	0.52
s_{10}	9.70	$f_{9,10}$	10.60				

Figure 1. Spot Curve. Curve of One-Year Forward Rates and Implied Spot Curve One Year Forward

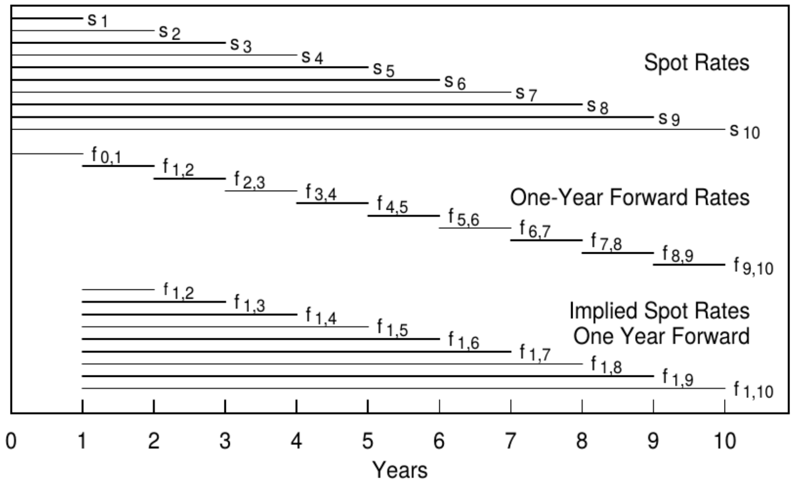

Figure 2. Future Years Covered by Each Rate

Forwards as Break-Even Rates

> Forward yield as break-even yield

The numbers in column D in Figure 1 —— the difference between the implied spot curve one year forward and today's spot curve —— show that "the forwards imply rising rates." How should this statement be interpreted? It does not necessarily mean that the market expects rising rates. Instead, the forwards tell how much the spot curve needs to change over the next year to make all bonds earn the same holding-period return. Recall that the holding-period return is a sum of a bond's initial yield and its capital gains or losses caused by yield changes. For example, if today's spot curve is upward sloping, longer-term bonds have a yield advantage over the one-period bond. To equate holding-period returns across bonds, longer bonds have to suffer capital losses that offset their initial yield advantage. Forwards show exactly how much long-term rates have to increase to cause such capital losses. Stated in terms of rate levels instead of rate changes, the implied spot rates one year forward (column C in Figure 1) are such future spot rates that would make all government bonds earn the same holding-period return over the next year. (Moreover, this same return must be the return of the one-year zero because it is already known today.) This break-even relation follows from Equation (3) by setting m = 1 and rearranging:

> The numbers in column D of Figure 1—the difference between the implied spot yield curve one year from now and the current spot yield curve—suggest that “forward yields imply an increase in yields.” How should this statement be interpreted? This does not necessarily mean that the market expects yields to rise. Rather, forward yields tell us how much the spot yield curve needs to change next year to give all bonds the same holding period return. Recall that the holding period return is the sum of a bond’s initial yield and the capital gains or losses resulting from changes in its yield. For example, if the current spot yield curve is upward sloping, long-term bonds have a yield advantage relative to one-year bonds. To make the holding period returns equal, longer-term bonds must suffer capital losses to offset their initial yield advantage. Forward yields show how much the long-term yield must increase to cause this capital loss. Based on the level of yield rather than changes in yield, the implied spot yield one year from now (column C in Figure 1) will give all government bonds the same holding period return next year. (Furthermore, this return must be the return of a one-year zero-coupon bond, as it is currently known.) By setting m = 1 and rearranging, we get from equation (3):

$$\frac{(1+s_n)^n}{(1+f_{1,n})^{n-1}} = 1 + s_1. \tag{4}$$

The left-hand side of Equation (4) is the return of buying an n-year zero at rate s_n today and selling it a year later at rate $f_{1,n}$. The right-hand side is the riskless return of the one-year zero. Thus, $f_{1,n}$ is the selling rate at which the n-year zero's holding-period return equals the return of the riskless asset. A numerical example illustrates the computation of the break-even rate $f_{1,2}$. The numbers are from Figure 1: a 6% one-year spot rate and a 7% two-year spot rate. Over the next year, the return of a one-year zero is known to be 6%, while the holding-period return of a two-year zero depends on its selling price at the end of the year —— when its remaining maturity is one year. So, the question is: “What should the one-year spot rate be one year hence to make the zero longer's holding-period return 6%?" A little math shows that the answer is 8.01%. At this selling rate, the longer zero's price would rise from 87.34 ($= 100/(1.07^2)$) to 92.58 ($= 100/1.0801$), earning it 6% ($= 92.58/87.34 - 1$). Thus, the implied one-year spot rate one year forward, $f_{1,2}$= 8.01%, is the level of future one-year rate that would make investors ex post indifferent to holding either of the two zeros.

> The left side of equation (4) is the current...s_nBuy an n-year zero-coupon bond at the yield and return it at the yield after one year.$f_{1,n}$The return generated from selling. The right side shows the risk-free return of a one-year zero-coupon bond. Therefore,$f_{1,n}$This is the selling yield that makes the return on an n-year zero-coupon bond equal to the risk-free rate of return. An example illustrates the break-even yield.$f_{1,2}$The calculations are as follows. The figures are from Figure 1: 6% is the one-year spot yield, and 7% is the two-year spot yield. In the following year, the yield on a one-year zero-coupon bond is 6%, while the yield on a two-year zero-coupon bond depends on the price at the end of the year. So, the question is: "What should the spot yield be one year from now so that the return on holding a longer-term zero-coupon bond equals 6%?" Mathematical calculations show the answer is 8.01%. Selling at this yield, the price of the longer-term zero-coupon bond will be from 87.34 (...). $= 100/(1.07^2)$The index rose to 92.58.$= 100/1.0801$), and receive a 6% return ($= 92.58/87.34 - 1$Therefore, the implied one-year spot yield one year from now...$f_{1,2}$= 8.01%. If the spot yield reaches this level in the next year, it will give investors the same return on two zero-coupon bonds.

It is worth reiterating that the forwards imply distinct yield changes for actual bonds and for constant-maturity rates. The forwards tell how much the yield of a given security — a longer-term bond — needs to change to offset an initial yield spread over the short-term rate. Alternatively, the forwards tell how much a given point on the spot curve has to shift to equate holding-period returns across bonds. In the example above, the two-year zero's yield had to rise by 1.01% and the constant-maturity one-year spot rate had to rise by 2.01%. The 1% difference is due to the so-called “rolling down the yield curve” effect, which we discuss shortly. However, first, we provide a rule-of-thumb relation between initial yield spreads and break-even yield changes —— and some economic intuition about this relation. We compute the implied two-year spot rate one year forward ($f_{1,3}$) by equating the holding-period returns of the three-year zero and the one-year zero over the next year. (Recall that the value of the two-year spot rate after one year determines the holding-period return of what is today a three-year zero.) The left-hand side of Equation (5) is the approximate holding-period return of the three-year zero [4] (given a selling rate $f_{1,3}$), and the right-hand side is the holding-period return of the one-year zero.

> It is worth reiterating that forward yields imply different yield changes for different bonds and maturities. Forward yields tell us how much the yield on a given bond (a long-term bond) needs to change to offset the initial spread relative to short-term yields. Alternatively, forward yields determine how much the spot yield curve must shift to equalize the holding period returns for different bonds. In the example above, the yield on a two-year zero-coupon bond must rise by 1.01%, and the one-year spot yield must rise by 2.01%. This 1% difference is due to the so-called "yield curve deflection effect," which we will discuss later. We provide an empirical relationship between the initial spread and the change in break-even yield, along with some economic intuition about this relationship. We calculate the implied two-year spot yield one year later by making the holding period return of a three-year zero-coupon bond equal to the return of a one-year zero-coupon bond in the following year. $f_{1,3}$ (Recall that the value of the two-year spot yield one year later determines the holding period return of the current three-year zero-coupon bond.) The left side of equation (5) is the approximate holding period return of the three-year zero-coupon bond (given the selling yield). $f_{1,3}$ The right side shows the holding period return of a one-year zero-coupon bond.

$$s_3 - Dur_2 * (f_{1,3} - s_3) \approx s_1. \tag{5}$$

Note that $f_{1,3} - s_3$ is not the three-year zero's actual subsequent yield change but the break-even yield change implied by the forward rates today. Rearranging Equation (5) gives a rule-of-thumb that the break-even yield change for the three-year zero equals the yield spread divided by the bond's duration (at horizon): $f_{1,3} - s_3 \approx (s_3 - s_1)/Dur_2$. The following observation may be helpful. A large yield spread means that a purchase of a three-year zero financed by the sale of a one-year zero has a large positive “carry.” The profit of this position is the sum of the yield carry ($s_3 - s_1$) and the capital gains or losses caused by the longer zero's yield changes. The position is bullish —— it profits from falling rates and suffers from rising rates —— but the positive carry provides a cushion against rising rates. The trade will lose money only if the two-year spot rate rises above $f_{1,3}$ in one year.

> Notice, $f_{1,3} - s_3$ It is not the subsequent actual yield change of a three-year zero-coupon bond, but rather the change in the break-even yield implied by the current forward yield. Rearranging equation (5) gives a rule of thumb that the change in the break-even yield of a three-year zero-coupon bond equals the yield spread divided by the bond's duration: $f_{1,3} - s_3 \approx (s_3 - s_1)/Dur_2$ The following observation may be helpful: a large interest rate spread means that shorting one-year zero-coupon bonds and going long three-year zero-coupon bonds has a large positive carry. The profit of this position is the carry ($s_3 - s_1$ This is the sum of the capital gains and losses resulting from changes in the yield of long-term zero-coupon bonds. This position is bullish—profiting from falling yields and losing from rising yields—but a positive carry acts as a buffer against rising yields, only becoming stronger if the two-year spot yield rises above [a certain level] within a year. $f_{1,3}$ That's when losses will occur.

The break-even yield change $f_{1,3} - s_3$ shows how much the three-year zero's yield can rise before its carry advantage is offset. However, if an upward-sloping yield curve remains unchanged, the zero's yield will fall as it rolls down the curve (from s_3 to s_2). The capital gains from this rolldown tendency provide the three-year zero with an additional cushion against rising rates. The yield advantage component and the rolldown component can be added up to answer the question, “How much should the constant-maturity two-year spot rate change in the next year to make the three-year zero and the one-year zero earn the same return?” The answer is the implied, or break-even, change in the two-year spot rate over the next year (Δf_2 (in column D in Figure 1):

> Changes in break-even rate of return $f_{1,3} - s_3$ This shows how much the yield on a three-year zero-coupon bond can rise before its spread advantage is offset. However, if the upward-sloping yield curve remains unchanged, the yield on the zero-coupon bond will fall along the curve (from...). s_3 arrive s_2 This downward trend in capital returns provides an additional buffer against rising yields for three-year zero-coupon bonds. The yield advantage and the yield decline together answer the question: how much does the two-year spot yield need to change one year later for a three-year zero-coupon bond to have the same return as a one-year zero-coupon bond? The answer is the implied or break-even change in the two-year spot yield the following year (column D in Figure 1). Δf_2):

$$\Delta f_2 = (f_{1,3} - s_3) + (s_3 - s_2) = f_{1,3} - s_2. \tag{6}$$

If only the spot rates in Figure 1 are known (s_1= 6%,s_2= 7%,s_3= 7.75%), the yield advantage component is 0.875% ($= (7.75 - 6.00)/2$) and the rolldown component is 0.75% ($= 7.75 - 7.00$Thus,$\Delta f_2 = 1.625\%$($= 0.875 + 0.75$) and $f_{1,3} = 8.625\%$ ($= 7.00 + 1.625$In words,$\Delta f_2$is the difference between the implied two-year spot rate one year forward and the two-year spot rate today. Figure 3 illustrates graphically how this break-even yield change is decomposed into the break-even yield change needed to offset the carry ($f_{1,3} - s_3$) and the rolldown yield change ($s_3 - s_2$5

> If only the spot rate of return in Figure 1 is known (s_1= 6%,s_2= 7%,s_3= 7.75%), then the advantage of return is 0.875% ($= (7.75 - 6.00)/2$And the decline in yield was 0.75% ($= 7.75 - 7.00$).therefore,$\Delta f_2 = 1.625\%$($= 0.875 + 0.75$),$f_{1,3} = 8.625\%$ ($= 7.00 + 1.625$).in other words,Δf_2It is the difference between the two-year spot yield one year from now and the current two-year spot yield. Figure 3 shows how to decompose this change in break-even yield into offsetting Carry ($f_{1,3} - s_3$The portion of) and the portion of declining yield ($s_3 - s_2$).

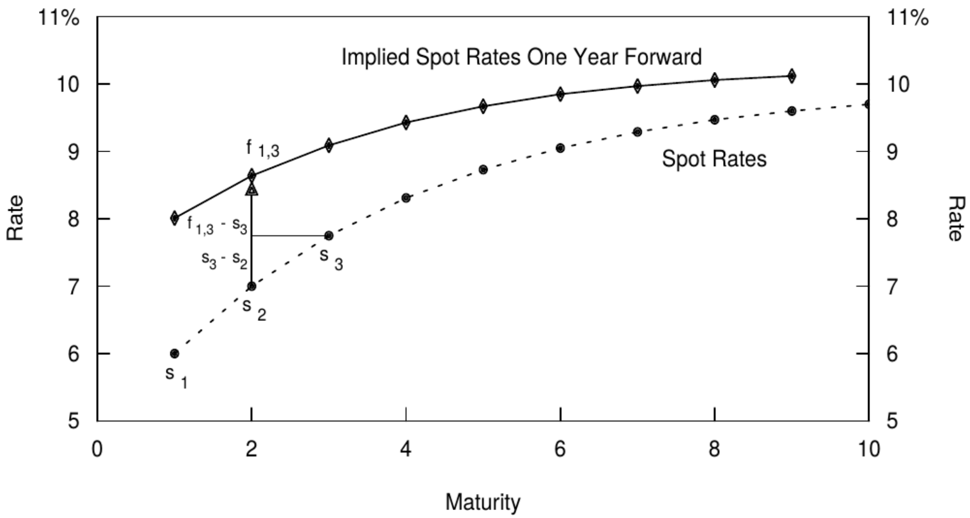

Figure 3. Spot Curve and Implied Spot Curve One Year Forward

An interesting relation exists between the curve of one-year forward rates and the implied spot curve one year forward (columns B and C in Figure 1). The steeper the former curve is, the higher the latter curve must be. A steep curve of one-year forward rates reflects a large rolling yield advantage for long-term bonds over the one-year bond. (We show in the Appendix that the one-year forward rate between maturities n-1 and n is equal to the n-year zero's rolling yield, that is, its one-year horizon return if the yield curve does not change. Thus, the one-year forward rate $f_{n-1,n}$is equal to the spot rate$s_n$plus the rolldown return.) The larger this rolling yield advantage is, the larger the yield increase required to offset it is, and the higher the implied spot curve one year forward is (see the Appendix for details).

> There is an interesting relationship between the one-year forward yield curve and the implied spot yield curve one year from now (columns B and C in Figure 1). The steeper the former curve, the higher the latter must be. A steep one-year forward yield curve reflects the significant rolling yield advantage of long-term bonds relative to one-year bonds. (As shown in the appendix, the one-year forward yield between n-1 years and n years is equal to the rolling yield of an n-year zero-coupon bond, which is the return over a one-year holding period assuming the yield curve remains unchanged. Therefore, the one-year forward yield...)$f_{n-1,n}$Equal to current yields_n(Including the decline return.) The greater the rolling yield advantage, the greater the increase in yield required to offset it, and the higher the implied spot yield curve one year later (see Appendix for details).

Break-Even Yield Changes for Curve-flattening Positions

> Changes in break-even return of flat curve positions

We can extend the above analysis to more complex yield curve positions. If the implied spot rates one year forward are realized, all self-financed positions of government bonds will break even (earn a return of 0%). To break even, any position with a positive carry will have to suffer capital losses; the forwards show how large spread changes over the next period would cause capital losses that offset the positive carry. Conversely, any position with a negative carry will have to earn capital gains to break even; thus, forwards imply spread changes that cause such capital gains. The following example clarifies this point.

> We can extend the above analysis to more complex yield curve trading. If the implied spot yield one year from now materializes, all self-financing Treasury positions will break even (earn a 0% return). To break even, any position with a positive carry must suffer a capital loss; the forward yield indicates how much the spread needs to change in the next period (year) for the resulting capital loss to offset the positive carry. Conversely, any position with a negative carry must generate a capital return to break even. Therefore, changes in the spread

implied by the forward yield lead to changes in capital returns. The following example illustrates this.

Consider an investor who has a strong view, with a one-year horizon, that the spot curve will flatten between two- and four-year maturities. He could implement this curve flattening view by selling a three-year zero and by buying with the sale proceeds equal market values of a five-year zero and a one-year zero (which represents cash at horizon). [6] Such a "barbell-bullet" trade is duration-neutral; thus, the position is not sensitive to parallel changes in interest rates, but it profits from the curve flattening. In a typical yield curve environment, this trade earns a negative carry. That is, when the spot curve is concave (steeper slope in the front end than in the long end), the yield loss of moving from the three-year zero to the one-year zero will be greater than the yield gain of moving from the three-year zero to the five-year zero. For example, in Figure 1 the negative carry is 39 basis points $(0.5 * (6.00 - 7.75) + 0.5 * (8.73 - 7.75) = -0.88 + 0.49 = -0.39)$. For the trade to make money, capital gains caused by future flattening of the spot curve must offset the negative carry.

> Suppose an investor strongly anticipates that the spot yield curve between two and four years will flatten after one year. He can implement this expectation by selling a three-year zero-coupon bond and buying an equal market value of five-year and one-year zero-coupon bonds (representing cash at maturity). This "barbell-bullet" trade is duration-neutral, therefore the position is insensitive to parallel changes in yields, but it profits from a flattening curve. In the case of a typical yield curve, this trade yields a negative carry. When the spot yield curve is convex (steeper at the short end than the long end), the yield loss of the three-year zero-coupon bond relative to the one-year zero-coupon bond will be greater than the yield gain of the three-year zero-coupon bond relative to the five-year zero-coupon bond. For example, in Figure 1, the negative carry is 39 basis points (...). $0.5 * (6.00 - 7.75) + 0.5 * (8.73 - 7.75) = -0.88 + 0.49 = -0.39$ To ensure that a trade is profitable, the capital returns generated by a flattened future spot yield curve must offset the negative carry.

The implied spot curve one year forward indicates the future level of the two- to four-year spread at which the trade —— like any bond position with no net investment —— exactly breaks even. This is the sense in which the forward rates "imply" flattening of the spot curve. (which is needed to offset the negative carry). If today's spot curve were linear and not at all curved, the flattening trade would give up no yield, and, consequently, the forwards would imply no flattening of the spot curve. The break-even change in the two- to four-year spread would be zero.

> The implied spot yield curve one year from now shows the future two-year and four-year yield spreads at which the trade (like any bond position with no net investment) breaks even. This is what it means for forward yields to "imply" a flattening of the spot yield curve. The greater the curvature of the current spot yield curve (for a given three-year yield, the one-year or five-year yield is relatively lower), the less attractive the flattening trade is (a larger negative carry), and the greater the implied flattening of the forward yield curve (to offset the negative carry). If the current spot yield curve were linear and completely flat, the flattening trade would not generate a negative carry; therefore, the forward yield implies that the spot yield curve would not flatten. The break-even change in the two-year to four-year yield spread would be zero. Finally, if the current spot yield curve were convex downwards, the (barbell-bullet) flattening trade would immediately generate a positive carry, and the forward yield would imply that the spot yield curve would become steeper.

Note that the break-even change in the (constant-maturity) two- to four-year yield spread depends not only on the barbell's and the bullet's initial yields, but also on their rolldown tendencies. When the spot curve is concave, a rolldown return advantage augments the bullet's yield advantage. (The three-year zero rolls down a steeper part of the curve than the five-year zero, and the one-year zero earns no rolldown return because it has zero duration at horizon.) Thus, the rolling yield curve —— that is, the curve of one-year forward rates —— is even more concave than the spot curve. The amount of implied flattening (which is needed to offset the negative yield carry and the difference in rolldown returns) really depends on the curvature of the rolling yield curve.

> Note that the break-even change in the two- to four-year yield spread depends not only on the initial yields of the barbell and bullet portfolios but also on their yield decline trend. When the spot yield curve is convex, the decline return amplifies the yield advantage of the bullet portfolio. (Compared to a five-year zero-coupon bond, the three-year zero-coupon bond declines in a steeper part of the curve, while a one-year zero-coupon bond has no decline return because its duration is zero at maturity.) Therefore, the rolling yield curve (the curve of one-year forward yields) is more convex than the spot yield curve. The implied flattening (to offset the difference between negative carry and decline returns) actually depends on the curvature of the rolling yield curve.

Figure 4 illustrates a typical spot curve and the corresponding curve of one-year forward rates (columns A and B in Figure 1). To evaluate the degree of curvature in these curves, we draw a straight line between a pair of zeros with different maturities (durations). Each point on these lines represents a barbell portfolio of the two zeros. The curvature is measured by the vertical distance between a barbell portfolio and a duration-matched bullet bond. The curvature of the spot curve reflects the bullet-barbell yield spread, while the curvature of the curve of one-year forward rates reflects the bullet-barbell rolling yield differential. The larger the vertical distance between the barbell and the duration-matched bullet is, the more “expensive” the barbell is in the sense that larger curve flattening is needed for the trade to break even. For example, the initial rolling yield differential of 105 basis points at the three-year point ($0.5 * 6.00 + 0.5 * 10.44 - 9.27 = -1.05$) will only be offset if the two- to four-year spread narrows by 52 basis points in one year. The break-even spread change can be computed from column D in Figure 1:$112 - 164 = -52$basis points.

Figure 4 shows a typical spot yield curve and its corresponding one-year forward yield curve (columns A and B in Figure 1). To calculate the curvature in these curves, we draw a straight line between a pair of zero-coupon bonds with different maturities (durations). Each point on these lines represents a barbell combination of two zero-coupon bonds. Curvature is measured by the vertical distance between the barbell combination and the duration-matched bullet combination. The curvature of the spot yield curve reflects the spread of the bullet-barbell combination, while the curvature of the one-year forward yield curve reflects the rolling yield difference of the bullet-barbell combination. The greater the vertical distance between the barbell combination and the duration-matched bullet combination, the more "expensive" the barbell combination is, because this means a greater degree of curve flattening is required for the trade to break even. For example, the initial rolling yield difference for a three-year term is 105 basis points (…). $0.5 * 6.00 + 0.5 * 10.44 - 9.27 = -1.05$The break-even spread will only be offset if the spread between the two-year and four-year maturities narrows by 52 basis points within a year. The change in the break-even spread can be calculated from column D in Figure 1. $112 - 164 = -52$One base point.

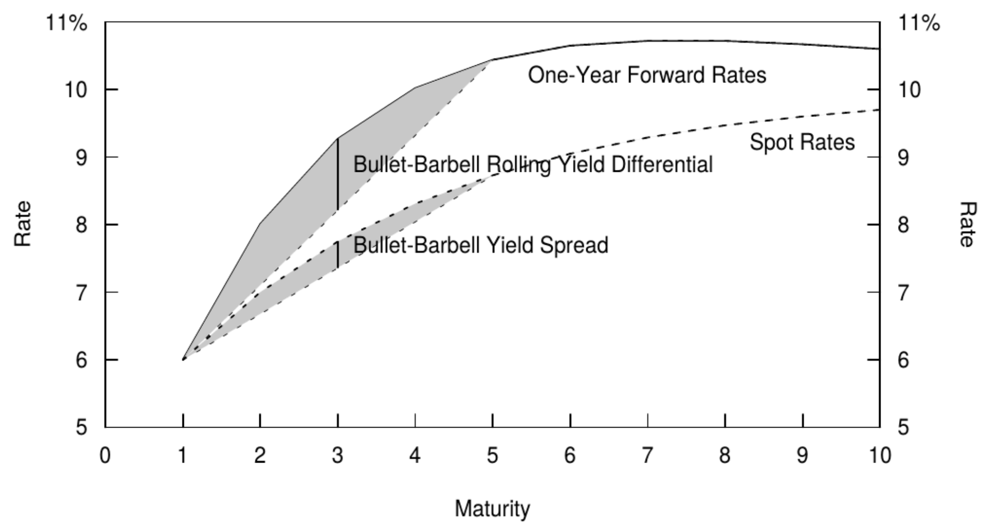

Figure 4. Measuring the Degree of Curvature

THE EXPECTATIONS HYPOTHESIS AND THE YIELD CURVE

Expectations Hypothesis and Yield Curve

How Do Rate Expectations Influence Today's Yield Curve Shape?

How do yield expectations affect the current shape of the yield curve?

It is widely agreed that the market's rate expectations have a strong influence on the yield curve shape. It is much more controversial to argue that such expectations are the only determinant of the yield curve shape. [7] However, this is roughly what the pure expectations hypothesis (PEH) claims. This hypothesis assumes that all government bonds, regardless of their maturity, have the same near-term expected return. The motivation is that the market prices of bonds are set by risk-neutral traders, whose activity eliminates any expected return differentials across bonds.

It is well known that market yield expectations have a significant impact on the shape of the yield curve. However, it is controversial that these expectations are the sole determinant of the yield curve's shape. The Perfect Expectations Hypothesis (PEH) holds this view. This

> hypothesis assumes that all government bonds, regardless of their maturity, have the same near-term expected return. The motivation for this hypothesis is that bond market prices are set by risk-neutral traders whose trading activity eliminates differences in expected returns between different bonds.

If all government bonds have the same near-term expected return, any yield differences across bonds must imply expectations of future rate changes (so that expected capital gains or losses offset the impact of initial yield differences). For example, if investors expect rates to rise and long-term bonds to lose value, they require higher initial yields for long-term bonds than for short-term bonds, making today's yield curve upward sloping. This kind of break-even argument is similar to the one used in the previous section, except that now the expected (as opposed to realized) returns are being equalized across bonds.

> If all government bonds have the same near-term expected return, any yield differences between different bonds imply expectations of future yield changes (and thus expected capital gains offset the effects of initial yield differences). For example, if investors expect yields to rise, the value of long-term bonds will fall, and the initial yield of long-term bonds is higher than that of short-term bonds, thus causing the current yield curve to slope upward. This break-even argument is similar to that used in the previous section, except that the expected (not realized) returns need to remain equal across different bonds.

Figure 5 illustrates how different types of expectations influence today's spot curve (if there are no expected return differences across bonds and if the convexity bias is ignored). Expectations of unchanged future rates lead to a horizontal spot curve, rising rate expectations (for example, a parallel shift of 100 basis points over one year) lead to a linearly upward-sloping spot curve and curve-flattening expectations lead to a concave curve shape. A break-even type of argument can motivate each situation:

> Figure 5 illustrates how different types of expectations affect the current spot yield curve (assuming no difference in expected returns between bonds and ignoring convexity bias). Expectations of constant future yields result in a horizontal spot yield curve, rising expected yields (e.g., an annual increase of 100 basis points) result in a linearly upward-sloping spot yield curve, and a flattening expectation curve results in an upward-convex shape. A break-even argument can explain each of the following cases:

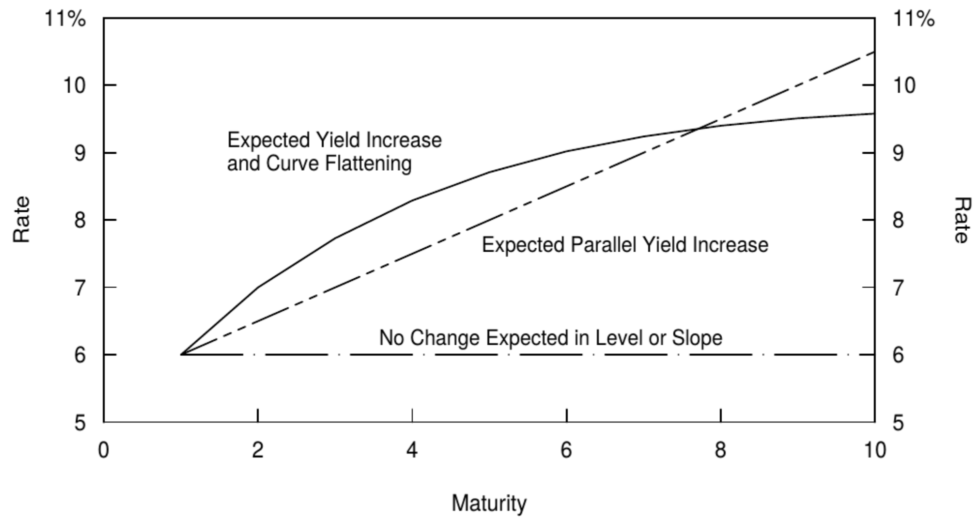

Figure 5. Spot Curves Given the Market's Various Rate Expectation

- If the market expects no rate changes, today's spot curve is flat because no expected gains or losses need to be offset by an initial yield spread.

> - If market expectations of returns remain unchanged, the current spot yield curve would be flat, as there would be no expected gains or losses to be offset by the initial spread.

- If the market expects rates to rise in a parallel fashion, longer-term bonds are expected to earn greater capital losses than shorter-term bonds. An initial yield advantage must offset these expected losses. Because the expected capital losses are proportional to duration, the yield advantage is also proportional to duration. Therefore, today's spot curve is linearly upward sloping. In a similar way, expectations of declining future rates make today's spot curve inverted.

> - If market expectations of rising yields simultaneously lead to greater losses for long-term bonds than for short-term bonds, the initial yield advantage must offset these expected losses. Since expected losses are proportional to duration, the yield advantage is also proportional to duration. Therefore, the current spot yield curve is linearly upward sloping. Similarly, expectations of future yield declines cause the current spot yield curve to invert.

- If the market expects the curve to flatten in the future, barbells and other curve-flattening positions are expected to earn capital gains. An initial negative carry must offset these expected capital gains. Therefore, today's spot curve is concave, and barbell portfolios have lower yields and rolling yields than duration-matched bullet bonds. In a similar way, expectations of future curve steepening tend to make today's spot curve convex, and barbells have higher yields than bullets.

> - If the market anticipates a flattening yield curve in the future, barbell portfolios and other curve-flattening positions are expected to yield returns. The initial negative carry must offset these anticipated returns. Therefore, the current spot yield curve is convex upwards, and the yield and rollover yield of barbell portfolios are lower than those of duration-matched bullet portfolios. Similarly, expectations of a steepening yield curve in the future tend to cause the current spot yield curve to convex downwards, and the yield of barbell portfolios is higher than that of bullet portfolios.

Expectations versus Risk Premia

> Expectations and risk premium

We emphasize that the PEH is nothing more than a hypothesis. Much empirical evidence shows that the extreme assumption of equal expected returns across bonds is false. Thus, it is unreasonable to assume that an upward-sloping yield curve reflects only expectations of rising rates. It is at least as reasonable to assume that such a shape reflects only the premium that investors require for holding long-term bonds. (We ignore the convexity bias until the next subsection.) In this light, the numbers in Figure 1 (s_1= 6%,s_2= 7%,$f_{1,2}$= 8.01%) can be interpreted in two ways.

> We must emphasize that the PEH is merely a hypothesis. Numerous empirical evidences suggest that assuming equal expected returns across bonds is flawed. Therefore, it is unreasonable to assume that an upward-sloping yield curve merely reflects expectations of rising yields. At the very least, such a shape can be considered to reflect only the premium demanded by investors holding longer-term bonds. (We ignore the convexity bias mentioned in the next unit). At this point, the figures in Figure 1 (s_1= 6%,s_2= 7%,$f_{1,2}$= 8.01%) can be interpreted in two ways.

- According to the PEH, the one- and two-year zeros have the same expected return over the next year. The return of the shorter zero is known to be 6%. The one-year return of the longer zero will be 6% only if its yield rises to 8.01% (thereby causing capital losses that offset its initial yield advantage). Thus,$f_{1,2}$reflects the expected level of the one-year spot rate one year hence.

> - According to PEH, one-year and two-year zero-coupon bonds have the same expected return in the following year. The yield on shorter-term zero-coupon bonds is known to be 6%. Only if their yield rises to 8.01% (thus causing capital losses to offset their initial yield advantage) will the yield on longer-term zero-coupon bonds be 6% in the following year. Therefore,$f_{1,2}$It reflects the expected level of the one-year spot yield one year from now.

- According to the risk premium hypothesis,$f_{1,2}$reflects only the longer zero's one-year expected return and no expected rate changes. Recall that one-year forward rates measure the zeros' one-year expected returns given no change in the yield curve. If the spot curve is unchanged in a year, the longer zero earns the initial 7% yield plus a 1.01% rolldown return when its yield declines to 6%.

> - According to the risk premium hypothesis$f_{1,2}$It only reflects the expected return of longer-term zero-coupon bonds one year from now, not the expected change in yield. Recall that, assuming the yield curve remains unchanged, the one-year forward yield measures the expected return of a zero-coupon bond one year from now. If the spot yield curve remains unchanged over a year, then when the yield on a zero-coupon bond falls to 6%, the longer-term zero-coupon bond receives the initial 7% yield plus a 1.01% decline return.

In both cases, the longer zero will earn the same return over a two-year period (14.49%); this return is known from its 7% annual yield today. However, in the first case the zero is expected to earn 6% in year one and 8.01% in year two, while in the second case, it is expected to earn 8.01% in year one and 6% in year two.

> In both scenarios, the longer-term zero-coupon bond will achieve the same return (14.49%) over the two-year period, derived from the current annualized yield of 7%. However, in the first scenario, the zero-coupon bond will achieve a 6% yield in the first year and an 8.01% yield in the second year; while in the second scenario, the zero-coupon bond will achieve an 8.01% yield in the first year and a 6% yield in the second year.

Let us put this example in a broader context. We show in the Appendix (Equation (13)) that, as a linear approximation, the yield change implied by the forwards can be split to an expected change in the n-1 year spot rate over the next year, $E(\Delta s_{n-1})$, and a bond risk premium (that is, the expected return of an n-year bond over the next year in excess of the riskless one-year rate, BRP_n):

> Let us broaden this example. As shown in the appendix (Equation (13)), as a linear approximation, the change in yield implied by the forward yield can be decomposed into the expected change in the spot yield in the next n-1 year. $E(\Delta s_{n-1})$ And the bond risk premium (i.e., the portion of the expected return of an n-year bond over the next year that exceeds the one-year risk-free rate). BRP_n):

$$f_{1,n} - s_{n-1} \approx E(\Delta s_{n-1}) + BRP_n/(n-1) \tag{7}$$

Equation (7) helps in contrasting different assumptions about the yield curve behavior. For better understanding, one can think of $f_{1,n} - s_{n-1}$ loosely as one measure of the yield curve steepness. Thus, the equation says that the curve steepness reflects market's future rate expectations, or expected return differentials across bonds, or some combination. The two cases above make two polar assumptions:

> Equation (7) helps to compare different assumptions about the behavior of the yield curve. For better understanding, one can... $f_{1,n} - s_{n-1}$ Roughly viewed as a measure of the steepness of the yield curve. Therefore, the equation suggests that the steepness of the curve reflects market expectations of future yields, or differences in expected returns between different bonds, or a combination thereof. The above two cases make two extreme assumptions:

- The PEH assumes that $BRP = 0$. Thus, all government bonds have the same near-term expected return as the riskless asset, and forwards reflect only the market's expectations of future rate changes.

> - PEH hypothesis $BRP = 0$ Therefore, as risk-free assets, all government bonds have the same expected near-term return, while forward yields only reflect market expectations of future yield changes.

- The risk premium hypothesis assumes that $E(\Delta s_{n-1}) = 0$. Thus, forwards reflect only the near-term expected return differentials across bonds.

> - Risk premium hypothesis assumes $E(\Delta s_{n-1}) = 0$ Therefore, forward yields only reflect the differences in near-term expected returns between different bonds.

In reality, of course, neither polar assumption is correct; the truth lies somewhere between. [8] Fortunately, the interpretation of forward rates as break-even rates is valid whether forward rates reflect the market's expectations of future rates, risk premia, or both. We will present some empirical evidence later that indicates that if one of the two polar assumptions has to be chosen, the risk premium hypothesis is the more realistic one.

> Of course, in reality, neither of these extreme assumptions is correct; the truth lies somewhere in between. Fortunately, interpreting forward yields as the break-even rate of return is reasonable, regardless of whether forward yields reflect market expectations of future returns, a risk premium, or both. We will present some empirical evidence later that, if necessary to choose between the two extreme assumptions, the risk premium hypothesis is generally closer to reality.

Alternative Versions of the Expectations Hypothesis and the Convexity Bias

> Alternative versions of the expected hypothesis and convexity deviation

The bond risk premium is not the only reason that the forward rates are not equal to expected spot rates. Another reason is the so-called convexity bias. The PEH is often identified with two statements: “All bonds have the same near-term expected return” and “forward rates are optimal (unbiased) forecasts of future spot rates.” It turns out that these two statements are not exactly consistent with each other versions. That is, two distinct versions of the PEH exist; the local expectations hypothesis is associated with the first statement and the unbiased expectations hypothesis is associated with the second statement. The difference between these hypotheses is related to convexity, that is, the nonlinearity in a bond's price-yield curve.

> Bond risk premium is not the only reason why forward yields do not equal expected spot yields. Another reason is the so-called convexity bias. The PEH (Premium for Expectations and Futures) is typically defined by two statements: all bonds have the same near-term expected return; and the forward yield is the best (unbiased) forecast of future spot yields. These two statements, it turns out, are not entirely consistent. That is, there are two different versions of the PEH; the partial expectations hypothesis is associated with the first statement, while the unbiased expectations hypothesis is associated with the second. The

difference between these hypotheses relates to convexity, the non-linear part of the relationship between bond prices and the yield curve.

We only try to give here some intuition about the convexity bias. Consider a situation in which the spot curve and the implied forward curves are flat at the 6% level. Because there are no yield differences across bonds and no rolldown, will the expected returns be equal across bonds? No, they will not, because some bonds are more convex than others. Positive convexity can only increase expected return for a given yield; thus, the longer-duration bonds, which exhibit greater convexity, will have higher expected returns. Therefore, even though forward rates equal expected spot rates in this example, all bonds will not have the same expected return. We refer to the impact of convexity on the yield curve shape as the convexity bias. [9]

We'll only attempt to provide some intuitive understanding of convexity bias. Consider a scenario where the spot yield curve and the implied forward yield curve are flat at the 6% level. Since there are no yield differentials between bonds, and no downside returns are generated, will the expected returns of all bonds be equal? No, because some bonds will be more convex than others. Positive convexity only increases the expected return for a given yield; therefore, longer-duration bonds with greater convexity will have higher expected returns. Thus, even if the forward yield is the same as the spot yield in this example, the expected returns of all bonds will not be the same. We call the effect of convexity on the shape of the yield curve convexity bias.

To conclude, in this section we have described how the market's rate expectations influence the shape of the yield curve, but we also emphasized that expectations are not the only determinants of the curve. The statement “forward rates show the market's expectations of future spot rates” is valid only if the bond risk premia and the convexity bias can be ignored. Ordinarily they cannot. In general, it is difficult to say whether the yield curve's upward slope reflects rising rate expectations or positive bond risk premia. It is equally difficult to say how much of the curvature in the yield curve reflects the market's flattening expectations and how much of it reflects the convexity bias. We will discuss these issues in the next section.

In summary, this section has described how market expectations of yields affect the shape of the yield curve, but we also emphasize that expectations are not the sole determinant of the curve. The statement that "forward yields reflect market expectations of future spot yields" is only valid if bond risk premiums and convexity bias are negligible, but typically these two factors are not. Generally, it is difficult to say whether the slope of the yield curve reflects expectations of rising yields or a positive bond risk premium. Similarly, it is difficult to say how much market expectation of flattening and convexity bias the curvature of the yield curve reflects. We will discuss these issues in the next section.

EMPIRICAL EVIDENCE ABOUT RATE EXPECTATIONS AND FORWARD RATES

Empirical results on the relationship between expected yield and forward yield

Do forwards reflect the market's rate expectations, required risk premia or both? Are forward rates or current spot rates better forecasts of future spot rates? Are expected bond returns equal across maturities, as the PEH asserts? In this final section, we address these questions and evaluate empirically what we really know about forward rates and the market's expectations. (We ignore nonlinear effects such as convexity bias and, thus, make no distinction between the different versions of the PEH.)

Do forward yields reflect market expectations of yields, the corresponding risk premium, or both? Which is a better predictor of future spot yields: forward yields or spot yields? Is it true, as PEH claims, that bonds of different maturities have equal expected returns? In this final section, we address these questions and assess our understanding of forward yields and market expectations based on real-world scenarios. (We ignore nonlinear effects such as convexity bias and therefore do not differentiate between different versions of PEH.)

Because expectations are not observable, academic researchers have studied these questions using two different methods. Many authors examine the forward rates' ability to predict actual subsequent rate changes and required bond risk premia. Others take a more direct approach and use surveys of interest rate forecasts to proxy for the market's rate expectations.

Because expectations are unobservable, academic researchers use two different approaches to study these issues. Many authors have investigated the predictive power of forward yields on actual subsequent yield changes and corresponding bond risk premiums. Others take a more direct approach, utilizing yield forecast surveys instead of market yield expectations.

Forwards' Ability to Forecast Future Rate Changes and Risk Premia

The predictive power of forward yields on future yield changes and risk premiums

We first examine forwards' ability to forecast future spot rate changes and realized future bond risk premia. The underlying idea is that the market's expectations are rational, and any forecast errors are “noise” that should wash out during the sample period. If the PEH holds, forwards are optimal predictors of future spot rates. (By “optimality,” we mean here unbiasedness. The forwards do not need to be very accurate predictors, but they should not contain systematic forecast biases.) However, if the risk premium hypothesis holds, forwards are optimal predictors of near-term expected bond returns, and the current spot curve is the optimal forecast for future spot curves. The implications of the two extreme hypotheses are summarized in Figure 6.

> We first examine the predictive power of forward yields on future changes in spot yields and the realization of future bond risk premiums. The basic idea is that market expectations are rational, and any prediction errors are simply "noise" that should be eliminated during the sampling period. If we adhere to the PEH hypothesis, then forward yields are the optimal predictor of future spot yields. (Here, "optimal" means unbiased; forward yields do not need to be highly accurate, but they should not contain systematic prediction biases.) However, if the risk premium hypothesis holds, forward yields are the best predictor of near-term expected returns on bonds, and the current spot yield curve is the best predictor of the future spot yield curve. The implications of these two extreme hypotheses are summarized in Figure 6.

	Pure Expectations Hypothesis	**Risk Premium Hypothesis**
What Is the Information in Forward Rates?	Market's Rate Expectations	Required Risk Premia
What Future Events Should Forward Rates Forecast?	Future Rate Changes	Near-Term Return Differentials Across Bonds
What is the Best Forecast of an n-Year Zero's One-Period Expected Return?	One-Period Riskless Rate	One-Period Forward Rate ($f_{n-1,n}$), that is, the Zero's Rolling Yield
What Is the Best Forecast of Next Period's Spot Curve?	Implied Spot Curve One Year Forward	Current Spot Curve
CORR(FSP_n, Δs_{n-1})	Positive	0
CORR(FSP_n, Realized BRP_n)	0	Positive

Figure 6. The Implication of Two Hypotheses About the Yield Curve Behavior

Figure 7 reports the correlation of the forward-spot premium ($f_{n-1,n} - s_1$orFSP_n; see Equation (16) in the Appendix), first, with the subsequent change in the n-1 year spot rate Δs_{n-1}over the next month and, second, with the subsequent realized bond risk premium (realizedBRP_nor the monthly holding-period return of an n-year zero in excess of the one-month rate). For better understanding, one can viewFSP_nas another measure of the yield curve steepness. We compute these correlations for six maturities (three- and six-month bills and estimated two-, three-, four-, and five-year zeros) using monthly Treasury market data from the 1970-94 period.

> Figure 7 reflects the forward-spot premium ($f_{n-1,n} - s_1$orFSP_nSee equation (16) in the appendix for its correlation, primarily the change in the spot yield over the next month after year n-1.Δs_{n-1}Second, the realized bond risk premium (realized)BRP_nOr the portion of the monthly holding yield of an n-year zero-coupon bond that exceeds the one-month yield. To better understand this, you can...FSP_nThis can be considered another measure of the steepness of the yield curve. We used monthly Treasury market data from 1970 to 1994 to calculate these correlations on six different maturities (three-month and six-month Treasury bills, and two-, three-, four-, and five-year zero-coupon bonds).

	3 Month	6 Month	2 Year	3Year	4 Year	5 Year
CORR(FSP_n, Δs_{n-1})	0.12	0.05	-0.05	-0.07	-0.11	-0.08
CORR(FSP_n, Realized BRP_n)	0.37	0.22	0.17	0.17	0.19	0.15

Source: Center of Research for Security Prices at the University of Chicago and Salomon Brothers Inc.

Figure 7. Evaluating Forward Rates' Ability to Predict Monthly Rate Changes and Risk Premia

The first row in Figure 7 provides the main finding. The forward-spot premia are negatively correlated with future changes in long-term rates. That is, when the yield curve is upward sloping, long-term rates do not tend to increase, as the PEH says they should, to offset their initial yield advantage over short-term bonds · Positively correlated with future bond risk premia (the second row). In the front end of the curve, forwards tend to at least predict the rate direction correctly. Therefore, the optimal forecast of a future spot rate is a weighted average of the current spot rate and the implied spot rate one period forward. In the long end, forwards appear to be inverse indicators of future rate changes. Overall, Figure 7 suggests that the yield curve steepness tends to reflect more near-term expected return differentials across bonds than the market's rate expectations. These findings are clearly inconsistent with the pure expectations hypothesis, but they may be explained by time-varying bond risk premia.

> The first row of Figure 7 shows the key findings. The forward-spot premium is negatively correlated with future changes in long-term yields. That is, when the yield curve slopes upward, long-term yields do not, as the PEH suggests, offset their initial yield advantage relative to short-term bonds with increased yields. Instead, long-term yields tend to decline, leading to increased capital returns and thus the yield advantage of long-term bonds.

Therefore, it is not surprising that the forward-spot premium is positively correlated with future bond risk premiums (second row). At the front of the curve, forward yields at least tend to correctly predict the direction of changes in spot yields. Therefore, the best predictor of future spot yields is a weighted average of current spot yields and implied spot yields. In the long run, forward yields appear to be an inverse indicator of future yield changes. Overall, Figure 7 suggests that the steepness of the yield curve tends to reflect more the differences in near-term expected returns between bonds than market expectations. These findings are clearly inconsistent with the perfect expectations hypothesis but may be explained by time-varying bond risk premiums.

We can relate our empirical analysis to the so-called persistence factors (PFs) that reflect the expectation that spot rates will remain at their current levels. [11] The two polar assumptions that we presented earlier — the PEH and the risk premium hypothesis — are associated with PFs 0 and 1. A zero value means that today's spot curve is expected to give way to the implied spot curve one period forward. A unit value means that today's spot curve is expected to remain unchanged (that is, to persist). Thus, if PF equals zero, forwards are optimal (unbiased) forecasts of future spot rates, and if PF equals one, current spot rates are the optimal forecasts. The PF can be estimated empirically by the slope coefficient in a regression of the annualized realized bond risk premium on the forward-spot premium (see Equation (16) in the Appendix). The value of the PF tells by how much an asset's near-term expected return increases for a given increase in the forward-spot premium. We find that for maturities beyond one year, for which forwards tend to give the wrong signal about future rate changes, the estimated PFs are greater than one (1.4 - 2.3). For maturities less than one year, for which forwards at least predict the rate direction correctly, the estimated PFs are smaller than one (about 0.8). These findings are consistent with the signs of the correlations in Figure 7.

We can link our empirical analysis to so-called persistence factors (PFs, reflecting expectations that spot yields will remain at their current levels). The two extreme hypotheses we mentioned earlier, the PEH and the risk premium hypothesis, have a 0-1 relationship with PF. 0 means that the current spot yield curve is expected to give way to the implied spot yield curve of the next period (year). 1 means that the current spot yield curve is expected to remain unchanged (i.e., persistent). Therefore, if PF equals 0, the forward yield is the best (unbiased) forecast of future spot yields; if PF equals 1, the current spot yield is the best forecast. PF can be empirically estimated by the slope coefficient of a linear regression of the realized annualized bond risk premium with respect to the forward-spot premium (see Equation (16) in the Appendix). The value of PF indicates how much the near-term expected return of an asset will increase with a given increase in the forward-spot premium. We found that for maturities longer than one year, with estimated PF values greater than 1 (1.4 to 2.3), forward yields tend to give false signals of future changes in spot yields. For maturities of less than one year, the estimated power factor (PF) is less than 1 (approximately 0.8), indicating that forward yields are at least correct in predicting the direction of change in spot yields. These findings are consistent with the sign of the correlation data in Figure 7.

Survey Evidence

Evidence from the investigation

A more direct approach is to use surveys of bond market analysts' interest rate forecasts to proxy for the market's rate expectations. The survey forecasts can be compared with forward rates; any difference should reflect a required risk premium.

A more direct approach is to use yield forecasts from bond market analysts instead of market yield expectations. These forecasts can be compared to forward yields, and any discrepancies should reflect the corresponding risk premium in the market.

We use the *Wall Street Journal*'s semiannual survey of Wall Street economists and analysts conducted 27 times between December 1981 and December 1994. In each survey, the participants predict the three-month bill rate and the 30-year bond yield at the end of the next June (December). We compute the expected rate change by subtracting the mid-December (June) rate from the survey median. [12] We then compare these survey forecasts with the yield changes that the forwards implies and with actual subsequent yield changes over each 6.5-month period. The forward yield changes are computed based on the on-the-run Treasury yield curve in mid-December (June). Figure 8 shows the main results separately for the six-month change in the bill rate and the bond rate and their spread.

We used 27 semi-annual surveys of Wall Street economists and analysts conducted by The Wall Street Journal between December 1981 and December 1994. In each survey, participants predicted the three-month Treasury discount rate and the 30-year bond yield at the end of the following June (December). We calculated the expected yield change by subtracting the mid-December (June) yield from the survey median. We then compared these survey forecasts with the yield change implied by forward yields and the actual yield change every 6.5 months. The forward yield change was calculated from the Treasury yield curve at mid-

December (June). Figure 8 shows the main results of the six-month changes in the discount rate and bond yields and their spreads, respectively.

A. Average Size of the Forecast Error (Mean Absolute Deviation)			
	Forward	**Survey**	**No Change**
Δ3-Month	1.21	1.03	1.03
Δ30-Year	0.88	1.00	0.86
ΔSpread (30 Year-3 Month)	0.91	0.75	0.81
B. Correlations Between Rate Changes			
	Forward/Survey	**Forward/Actual**	**Survey/Actual**
Δ3-Month	0.43	0.03	0.26
Δ30-Year	0.23	-0.01	-0.22
ΔSpread (30 Year-3 Month)	0.21	0.33	0.28
C. Average (Expected) Rate Change Over Six Months			
	Forward	**Survey**	**Actual**
Δ3-Month	+0.80	-0.26	-0.25
Δ30-Year	+0.10	-0.14	-0.24
ΔSpread (30 Year-3 Month)	-0.70	+0.12	+0.01

Figure 8. Accuracy and Bias of Implied Rate Predictions from Forwards and from a Wall Street Journal Survey

Panel A examines the accuracy of the forwards, the surveys and the no-change predictions by showing the average magnitude of the difference between the predicted yield change and the subsequent yield change. (Note that the no-change prediction is consistent with the risk premium hypothesis and that its forecast error is simply the actual yield change.) Panel A shows that all forecasts are quite poor; average errors are of the same order of magnitude as the average yield changes, roughly 100 basis points in six months. Another observation is that the forwards have somewhat larger forecast errors than the no-change predictions, suggesting that the current spot curve predicts future spot curves better than the implied forward curve does. Given the small number of observations, these differences are not statistically significant. A more serious problem regarding the forwards' predictive ability is that long-term rates tend to move away from the forwards, not toward them —— recall the evidence in Figure 7. However, the negative correlation is quite small in this sample (-0.01), and the expert forecasters have been even worse predictors of long-rate changes (the correlation is -0.22 in panel B). Panel C is the most interesting part of Figure 8. It reveals a systematic bias in the forward rates; during the 13-year period, the forwards typically imply rising short-term and long-term rates and a flattening yield curve, unlike the survey forecasts and the actual rate changes.

Table A examines the accuracy of forward yield forecasts, survey forecasts, and no-change forecasts by showing the average difference between predicted yield changes and subsequent yield changes. (Note that no-change forecasts are consistent with the risk premium hypothesis, and their forecast error is the actual yield change.) Table A shows that all forecasts are quite poor; the average error is the same as the average yield change, roughly 100 basis points over six months. Another observation is that forward yield forecasts have a larger forecast error than no-change forecasts, suggesting that the current spot yield curve is a better predictor of future spot yield curves than the implied forward yield curve. Given the small number of observations, these differences are not statistically significant. Recalling the evidence in Figure 7, a more serious problem with the predictive power of forward yields is that long-term yields tend to deviate from rather than converge to forward yields. However, the negative correlation in this sample is quite small (-0.01), and experts' predictions of long-term changes are even worse (correlation coefficient of -0.22 in Table B). Table C is the most interesting part of Figure 8. It reveals a systematic bias in forward yields; over a 13-year period, forward yields typically imply rising short- and long-term yields and a flattening yield curve, which differs from survey forecasts and actual yield changes.

Why are the forwards' implied rate predictions higher than survey forecasts and actual rate changes? A possible explanation is that the market requires a positive risk premium for holding the long-term bond. In fact, we can use the survey forecasts of future rates to compute a direct estimate of the average bond risk premium. The difference between the yield change implied by the forwards and the expected future yield change is approximately proportional to the risk premium. The 106 (80 - (-26))-basis point difference for the three-month bill translates to an average annualized risk premium of 53 basis points. [13] A similar calculation shows that the 24 (10 - (-14))-basis point difference for the 30-year bond translates to an average annualized risk premium of 480 basis points. These findings suggest that positive bond risk premia exist.

Why do forward yield forecasts show higher yields than both survey forecasts and actual yield changes? A possible explanation is that the market demands a positive risk premium for holding long-term bonds. In fact, we can use survey forecasts of future yields to calculate a direct estimate of the average bond risk premium. The difference between the yield change implied by forward yields and the expected change in future yields is roughly proportional to the risk premium. A difference of 106 (80 - (-26)) basis points in the discount rate of three-month Treasury bills translates to an average annualized risk premium of 53 basis points. Similar calculations show that a difference of 24 (10 - (-14)) basis points in the discount rate of

> 30-year bonds translates to an average annualized risk premium of 480 basis points. These findings suggest the existence of a positive bond risk premium.

Figure 9 shows how consistently the forward prediction of the three-month rate exceeds the survey forecast. Both series move together with the actual three-month bill rate (which is typically between the two series). As explained above, the difference between the two series is proportional to a bond risk premium. Figure 9 shows that this risk premium is not constant over time. Its time variation appears economically sensible in that the premium is exceptionally high around the 1982 recession and after the 1987 stock market crash when a recession was widely expected. The premium was quite low in 1993, despite the yield curve steepness, and it rose substantially in 1994. [14]

> Figure 9 shows that forward yield forecasts for the three-month yield consistently exceed survey forecasts. These two columns of forecasts move together with the actual three-month Treasury discount rate (which typically lies between the two series). As mentioned above, the difference between the two columns of forecasts is proportional to the bond risk premium. Figure 9 shows that this risk premium is not constant. Its time-varying nature seems economically reasonable, as the premium was very high during the 1982 recession and after the 1987 stock market crash when recessions were widely anticipated. Despite the steep yield curve, the premium was quite low in 1993 and rose sharply in 1994.

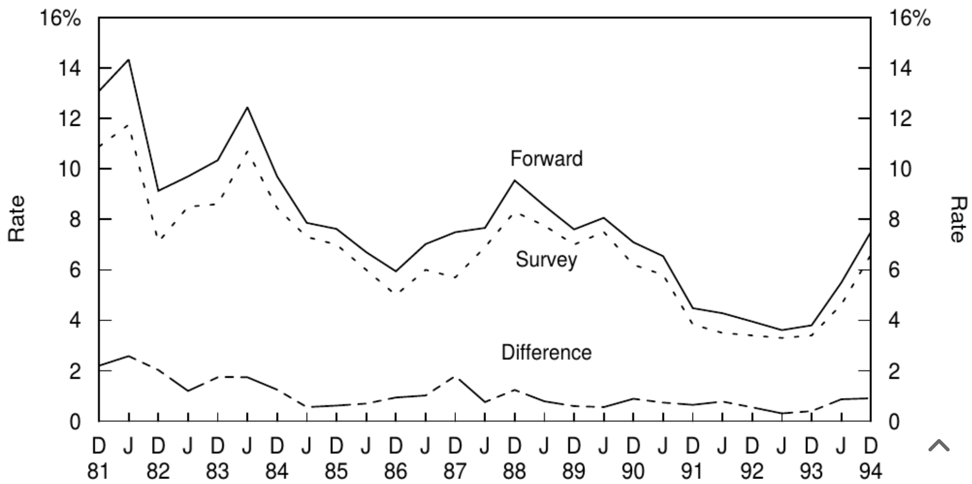

Figure 9. Forward- and Survey-Expected Three-Month Treasury Bill Rate Six Months Ahead

Figure 10 plots the implied forward changes in the three-month to 30-year yield spread and the corresponding survey forecasts. Forwards have quite consistently implied yield curve flattening, while the survey users were, until recently, often predicting curve steepening. We noted earlier that the typical concave yield curve shape causes forwards to imply curve flattening, and we conjectured that most of this implied flattening probably reflects the value of convexity rather than actual flattening expectations. Figure 10 is consistent with this conjecture. The difference between the two series in Figure 10 should, in fact, be a reasonable measure of the market price of convexity.

> Figure 10 shows the changes in the implied 3-month to 30-year yield spread by forward yields and the corresponding survey forecasts. Forward yields have fairly consistently implied a flattening yield curve, whereas until recently, survey results often predicted a steeper curve. We previously noted that the typical convex yield curve shape leads to a flattening implied by forward yields, and we hypothesized that most of this implied flattening likely reflects the value of convexity rather than actual expectations. Figure 10 is consistent with this conjecture. In fact, the difference between the two columns of forecasts in Figure 10 should be a reasonable measure of the market value of convexity.

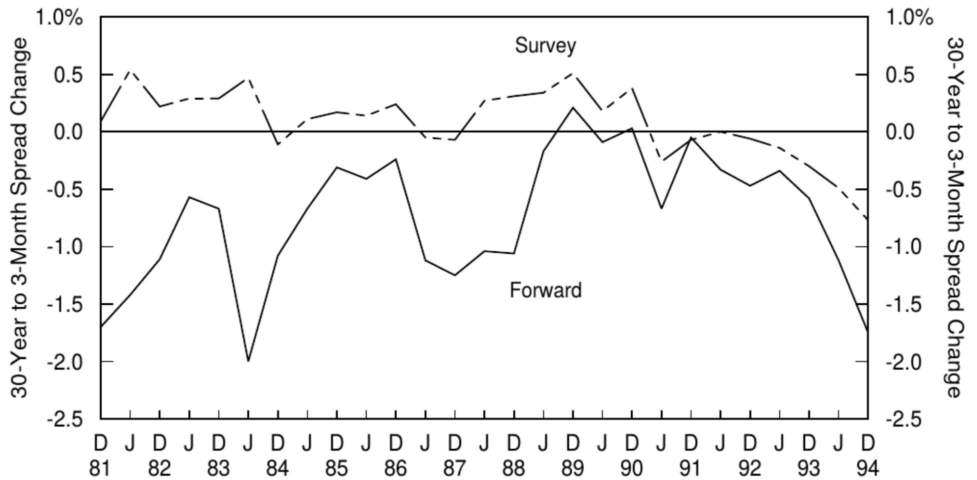

Figure 10. Forward- and Survey-Expected Yield Curve Steepening Six Months Ahead

Investment Implications of the Empirical Findings

> Empirical Results in Investment Practice

Overall, the empirical evidence is much more consistent with the risk premium hypothesis than with the pure expectations hypothesis. Forwards tell us more about near-term expected return differentials across bonds than about future rate changes. If either today's spot curve or the implied spot curve one period forward must be used as a predictor of the next period's spot curve, the evidence supports the use of the former as the neutral base case. These findings have obvious investment implications; rolling yield differentials between bonds or the corresponding break-even changes yield provide potentially useful relative value indicators for duration-extension trades. Intuitively, because the rate changes that forwards imply are not realized, on average, bonds with high rolling yields tend to keep their yield and roll down advantage. Similarly, break-even changes of yield spreads may be good relative value indicators for duration-neutral yield curve positions such as barbells versus bullets. In later reports, we will evaluate the historical performance of these indicators and traditional cheapness indicators such as yield spreads.

> Overall, empirical evidence aligns more closely with the risk premium hypothesis than with the full expectation hypothesis. Forward yields tell us more about the differences in near-term expected returns between different bonds than about future yield changes. If we must choose between the current spot yield curve or the implied spot yield curve as a forecast for the next period (year), empirical results support using the former as a neutral benchmark. These findings have significant investment implications; the difference in rolling yields between different bonds, or the corresponding change in break-even yields, provides potentially useful relative value indicators for trading strategies that increase duration. Intuitively, since the yield changes implied by forward yields are not realized, on average, bonds with higher rolling yields tend to maintain a yield advantage and a decline yield advantage. Similarly, the break-even change in spreads may be a good relative value indicator for duration-neutral positions (such as barbell-bullet combinations). In later reports, we will evaluate the historical performance of these indicators and traditional cheap indicators (such as spreads).

More generally, the empirical failure of the pure expectations hypothesis is good news for active managers. If all bond positions always had the same near-term expected returns, it would be extremely difficult to add value. Therefore, our findings provide some empirical justification for yield-seeking active investment strategies.

> More generally, the empirical failure of the perfect expectations hypothesis is good news for active managers. If all bond positions always have the same near-term expected returns, then adding value will be extremely difficult. Therefore, our findings provide some empirical evidence for seeking active investment strategies.

APPENDIX. RELATIONS BETWEEN FORWARD RATES, EXPECTED RATE CHANGES AND EXPECTED RETURNS

> Appendix: Relationship between forward yield, changes in expected yield, and expected return

This Appendix shows how forward rates are related to expected spot rate changes and expected holding-period returns and how one-year forward rates are related to implied spot rates one year forward. These relations are linear approximations that ignore nonlinear effects such as the convexity bias. Equation (3) shows how the annualized forward rate between maturities m and n is related to m- and n-year spot rates. By taking a first-order approximation of both sides of this equation and rearranging, we get a nice linear relation:

> This appendix shows how forward yields relate to changes in expected spot yields and expected holding period returns, and how one-year forward yields relate to implied spot yields one year later. These relationships are linear approximations that ignore nonlinear effects such as convexity bias. Equation (3) shows how annualized forward yields between maturities of m and n years relate to spot yields between m and n years. By approximating both sides of this equation with a first order and rearranging them, a good linear relationship is obtained:

$$f_{m,n} \approx \frac{ns_n - ms_m}{n-m}. \tag{8}$$

To compute the implied spot curve one year forward, we set m = 1:

> To calculate the implied immediate yield one year from now, we assume m = 1:

$$f_{1,n} \approx \frac{ns_n - s_1}{n-1}. \tag{9}$$

Equation (10) shows the n-year zero-coupon bond's holding period return over the next period (h_n). The zero earns its initial yield, s_n, plus a capital gain which is approximated by the product of the zero's duration at horizon (n-1) and its yield change.

> Equation (10) shows the return of the n-year zero-coupon bond held for the next period (year).h_nZero-coupon bonds obtain their initial yield.s_nIn addition, a capital gain is added, which is approximately the product of the duration (n-1) of a zero-coupon bond and the change in its yield.

$$h_n \approx s_n + (n-1) * (s_n - s^*_{n-1}), \qquad (10)$$

where$s^*$is the next period's rate (at which the bond is sold). Now we substitute Equation (10) into Equation (9), noting that$ns_n = s_n + (n-1) * s_n$:

> ins^*This is the yield for the next period (year) (bond sale period). Now we substitute equation (10) into equation (9) and note...$ns_n = s_n + (n-1) * s_n$:

$$f_{1,n} \approx \frac{h_n + (n-1) * s^*_{n-1} - s_1}{n-1} = s^*_{n-1} + \frac{h_n - s_1}{n-1}. \qquad (11)$$

We call$h_n - s_1$(the one-year holding-period return in excess of the short-term rate) the realized risk premium of the n-year zero. Because the equality in Equation (11) holds for realized returns, it also should hold in expectations if they are rational. Thus,

> We call$h_n - s_1$This is the risk premium realized by an n-year zero-coupon bond (the portion of the return over the one-year holding period that exceeds the short-term yield). Since the equation in (11) holds for the realized return, if the equation is reasonable, it should also apply to the expectation. Therefore,

$$f_{1,n} \approx E(s^*_{n-1}) + \frac{BRP_n}{n-1}, \qquad (12)$$

whereBRP_n(the bond risk premium) is the expected holding-period return of an n-year zero in excess of the riskless one-year rate or$E(h_n - s_1)$Subtracting$s_{n-1}$from both sides of Equation (12) gives the break-even change in the n-1 year spot rate:

> BRP_n(Bond risk premium) refers to the portion of the expected return of an n-year zero-coupon bond over its holding period that exceeds the one-year risk-free rate.$E(h_n - s_1)$ Subtract from both sides of equation (12)s_{n-1}The break-even change in the spot rate over a period of n-1 years was obtained:

$$\Delta f_{n-1} \equiv f_{1,n} - s_{n-1} \approx E(\Delta s_{n-1}) + \frac{BRP_n}{n-1}, \qquad (13)$$

where$\Delta s_{n-1} = s^*_{n-1} - s_{n-1}$(Note that)$\Delta s_{n-1}$denotes actual rate change over time, whereas$\Delta f_{n-1}$is the difference between two rates that are known today.) Equation (13) states that the break-even change in the n-1 year spot rate, implied by the forwards, is equal to the sum of the expected change in the n-1 year spot rate over the next year and the bond risk premium of the n-year zero (divided by n-1). The information in the forward rates reflects either expected future yield changes, or expected bond risk premia, or some combination of the two.

> in$\Delta s_{n-1} = s^*_{n-1} - s_{n-1}$(Notice,$\Delta s_{n-1}$This represents the change in the real rate of return, whileΔf_{n-1}(This is the difference between two currently known yields). Equation (13) shows that the break-even change in the n-1 year spot yield implied by the forward yield is equal to the sum of the expected change in the next n-1 year spot yield and the n-year zero-coupon bond risk premium (divided by n-1). The information in the forward yield reflects the expected future yield change or the expected bond risk premium, or some combination of both.

Similarly, we can compute an approximation of the one-year forward rates (m = n-1 in Equation (8)):

> Similarly, we can calculate an approximate value for the one-year forward yield (let m = n-1 in equation (8)):

$$f_{n-1,n} \approx ns_n - (n-1) * s_{n-1} = s_n + (n-1) * (s_n - s_{n-1}). \qquad (14)$$

Now we substitute Equation (10) into Equation (14):

> Now we substitute equation (10) into equation (14):

$$f_{n-1,n} \approx (n-1) * \Delta s_{n-1} + h_n. \qquad (15)$$

If the yield curve remains unchanged, the first term in the right-hand side of Equation (15) equals zero and$f_{n-1,n} = h_n$. In other words, the one-year forward rate is equal to a zero's holding-period return given an unchanged yield curve, or its rolling yield. Equation (14) shows that such return is equal to a sum of the initial yield and the rolldown return (the zero's duration at horizon (n-1) multiplied by the amount that the zero rolls down the yield curve as it ages).

If the yield curve remains unchanged, then the first term on the right-hand side of equation (15) equals zero and $f_{n-1,n} = h_n$ In other words, given a constant yield curve, the one-year forward yield is equal to the holding period return of a zero-coupon bond, or its rolling yield. Equation (14) shows that this yield is equal to the sum of the initial yield and the decline yield (the duration of the zero-coupon bond at the end of the holding period (n-1) multiplied by the amount the bond declines on the yield curve over time).

We can subtract s_1 from both sides of Equation (15) to get the forward-spot premium ($f_{n-1,n} - s_1$ or FSP_n and realized risk premium ($h_n - s_1$), and then take expectations:

We can subtract from both sides of equation (15) s_1 To obtain a forward-spot premium ($f_{n-1,n} - s_1$ or FSP_n) and realized risk premium ($h_n - s_1$),Then:

$$FSP_n \equiv f_{n-1,n} - s_1 \approx (n-1) * E(\Delta s_{n-1}) + BRP_n. \qquad (16)$$

A comparison of Equations (13) and (16) shows that the forward-spot premium is proportional to the break-even yield change: [15] $FSP_n = (n-1) * \Delta f_{n-1}$ Intuitively, FSP_n measures the rolling yield advantage of the n-year zero over the risky one-year zero, while Δf_{n-1} shows how large change in the n-1 year spot rate is needed to offset such a rolling yield advantage. The ratio of proportionality is n-1 because the product of duration-at-horizon and the break-even yield change gives a capital loss that is as large as the rolling yield advantage. Figure 11 illustrates this relation. The top panel shows a spot curve and the corresponding curve of one-year forward rates (rolling yields) which increase linearly with maturity. If the spot curve remains unchanged over the next year, the ten-year zero earns its annual yield 10.50% as well as a rolldown return of 4.50% (nine years end-of-horizon duration times 50 basis points rolldown yield change). Thus, its holding-period return is 15% if the spot curve does not change over the next year —— as shown by the right-most point on the curve of one-year forward rates. The ten-year zero has a 9% rolling yield advantage over the one-year zero. shows the implied spot curve one year forward (together with the same spot curve as in the top panel); that is, the break-even levels of future spot rates that would offset the longer-term bonds' rolling yield advantage over the one-year zero. The nine-year spot rate needs to increase by 100 basis points to cause a 9% capital loss for today's ten-year zero (whose maturity is nine years after a year). Similar calculations for various-maturity zeros show that a parallel increase of 100 basis points would make all government bonds earn the same 6% holding-period return as the one-year zero. (If convexity effects are taken into account, the break-even yield shift would not be exactly parallel.)

A comparison of equations (13) and (16) shows that the forward-spot premium is directly proportional to the change in the break-even rate of return: $FSP_n = (n-1) * \Delta f_{n-1}$ Intuitively, FSP_n The rolling yield advantage of an n-year zero-coupon bond compared to a risk-free one-year zero-coupon bond was measured, while Δf_{n-1} This shows how much the spot yield needs to change in n-1 years to offset this rolling yield advantage. The proportion is n-1 because the product of duration and break-even yield changes gives a capital loss as large as the rolling yield advantage. Figure 11 illustrates this relationship. The top figure shows a spot yield curve and a corresponding one-year forward yield (rolling yield) curve, growing linearly with maturity. If the spot yield curve remains unchanged in the next year, the annualized yield of a 10-year zero-coupon bond is 10.50%, with a decline yield of 4.50% (the yield decline change of 50 basis points over the nine-year duration at the end of the period). Therefore, if the spot yield curve does not change in the next year, its holding period return is 15%, as shown by the rightmost point on the one-year forward yield curve. The 10-year zero-coupon bond has a 9% rolling yield advantage over the one-year zero-coupon bond. The bottom figure shows the implied spot yield curve one year later (together with the same spot yield curve as the top figure); the future break-even level of the spot yield will offset the rolling yield advantage of the long-term bond over the one-year zero-coupon bond. A 100-basis-point increase in the nine-year spot yield would result in a 9% capital loss for a current ten-year zero-coupon bond (which will have a nine-year maturity in one year). Similar calculations for zero-coupon bonds across all maturities suggest that an average increase of 100 basis points would bring the holding-time return of all Treasury bonds to the same level as that of a one-year zero-coupon bond. (The change in break-even yield does not occur in a perfectly parallel manner if convexity effects are taken into account.)

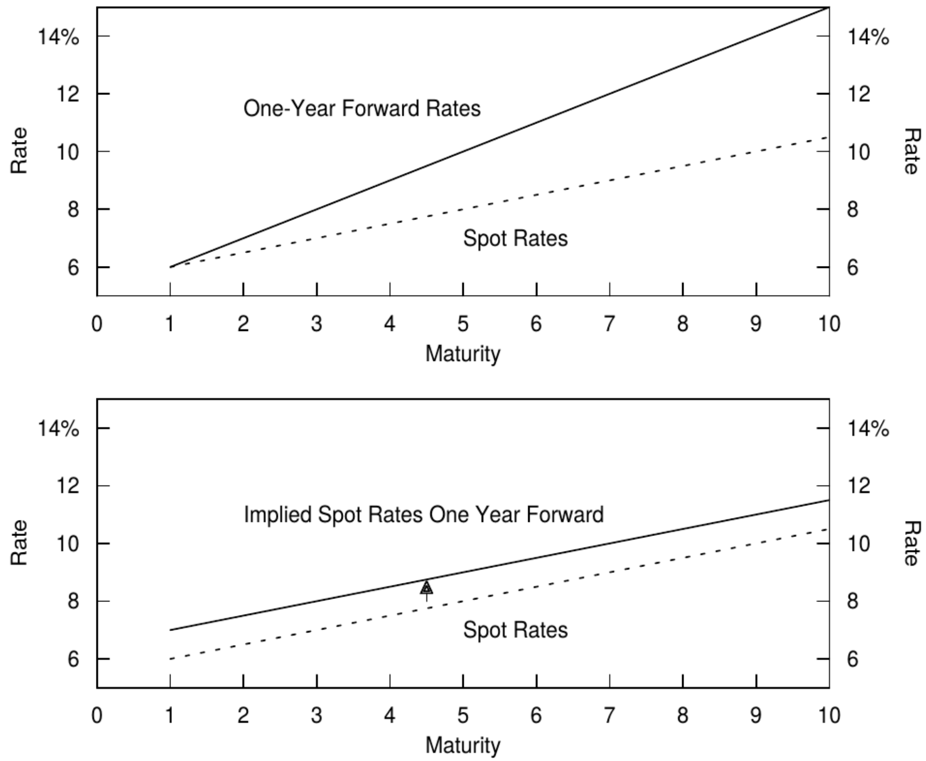

Figure 11. Link Between Zeros' Rolling Yields and Break-Even Spot Rate Changes

The link between initial rolling yields and the break-even changes in the spot rates is straightforward because the relationship is mathematical. If today's spot curve and the rolling yield curve were flatter than in the top panel in Figure 11, a smaller increase in the spot curve would be needed to offset the long-term bonds' rolling yield advantage over the one-year zero, and the forwards would imply smaller rate increases. If today's spot curve were inverted, long-term bonds would even have lower yields and rolling yields than the one-year zero, and the forwards would imply a rate decline to offset the long-term bonds' rolling yield disadvantage. If today's spot curve were upward sloping and concave (the “typical” shape; see Figure 4, which corresponds to the top panel in Figure 11), the forwards would imply rising rates and flattening curve to offset the rolling yield advantage of long-term bonds and steepening positions (bullets versus barbells; see Figure 3, which corresponds to the lower panel in Figure 11).

The relationship between the initial rolling yield and the break-even change in spot yield relies directly on mathematical derivation. If the current spot yield curve and rolling yield curve are flatter than those in Figure 11, the spot yield curve needs to rise slightly to offset the rolling yield advantage of long-term bonds compared to one-year zero-coupon bonds, while forward yields will imply a smaller yield increase. If the current spot yield curve is inverted, long-term bonds have lower yields and rolling yields compared to one-year zero-coupon bonds, while forward yields imply a decline in yields to offset the rolling yield disadvantage of long-term bonds. If the current curve is upward sloping and convex (a "typical" shape, see Figure 4, corresponding to the upper half of Figure 11), then forward yields will imply a rise in yields and a flattening curve to offset the rolling yield advantage of long-term bonds and steepening positions (bullet-barbell combinations, see Figure 3, corresponding to the lower half of Figure 11).

LITERATURE GUIDE

Literature Guide

Broad Surveys

Overview

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1. In practice, spot rates are rarely inferred from zero-coupon bond (STRIPS) prices because STRIPS differ from the more common Treasury coupon bonds in terms of liquidity and tax treatment. More often, the spot curve is estimated from the prices of all or most Treasury coupon bonds. Once the spot curve is known, it is a matter of algebra to construct the par yield curve which many investors prefer to follow. Alternatively, if the par curve is known, it is easy to construct the spot curve. For details, *see *Using STRIPS in a* Treasury Portfolio*, Janet Showers , Salomon Brothers Inc., August 1992, *and *Overview of Forward Rate Analysis*, Salomon Brothers Inc., May 1995. Similarly, if the coupon yield curve is known, it is easy to construct the spot yield curve. For detailed procedures, please refer to *Using STRIPS in a Treasury Portfolio* (Janet Showers, Salomon Brothers Inc., August 1992) and *Overview of Forward Rate Analysis* (Salomon Brothers Inc., May 1995)*
2. To simplify notation, this report focuses on spot rates rather than par yields. Furthermore, all analysis uses one-year forward rates and annual compounding frequency. *While semi-annual compounding and three-month forward rates are more common in practice, the equations would require including various annualization terms*
3. The forward yield path is computed by fixing m = n - 1 and letting it vary from 0 to 9. The implied spot yield curve one year forward is computed by fixing m = 1 and letting n vary from 2 to 10. In a similar way, we can construct the forward path of *any constant-maturity rate or the implied spot yield curve for any future horizon*
4. The holding-period return of a three-year zero over the next year is equal to its initial yield (s_3) plus the capital gains or losses caused by any yield change. In this example, we use linear approximation of the capital gains (percentage price change equals minus duration times yield change), ignoring convexity effects. For this reason, the forward rate $f_{1,3}$ That we compute will be 1.5 basis points lower than that in Figure 1 (which is computed without any approximation error). *The holding period return of the three-year zero-coupon bond in the following year is equal to its initial yield.s_3 This includes any capital gains or losses resulting from changes in yield. In this example, we use a linear approximation of the return on capital (percentage price change equals negative duration multiplied by yield change), ignoring the convexity effect. Therefore, we calculate the forward yield...$f_{1,3}$ This will be 1.5 basis points lower than in Figure 1 (without any approximation error calculation).*
5. The x-axis in Figure 3 (and in subsequent figures) is denoted as maturity, but it could also be denoted as *Macaulay duration because for zero-coupon bonds, duration equals maturity. (This report does not emphasize the distinction between Macaulay duration and modified duration.*
6. It might seem puzzling that a two- to four-year spread is achieved by trading three- and five-year zero-coupon bonds. The reason is that, in this example, the investor has a view on the spread changing within one year and the *zero-coupon maturities shortening by one year over the horizon. If the investor had a view with an immediate horizon, they would implement it by selling two-year bonds and buying four-year bonds, along with cash*
7. This report takes the market's rate expectations as given, except that this footnote briefly discusses the economic determinants of these expectations. An old Wall Street adage ——

that the central bank determines the level of short-term rates while the market's inflation expectations drive the long-term rates —— is an oversimplification, but probably captures the main determinants of rate behavior, (i) Market participants are likely to have clearer expectations regarding the near-term behavior of short rates than about more distant events. These expectations are closely related to the market's view on the economic conditions and on the direction of monetary policy, if the market expects an increase in economic growth rate and in credit demand, it often expects both the real rates and the inflation to increase, and vice versa. Moreover, central banks often try to influence inflation rates and economic growth by raising short-term (nominal and real) rates when fast economic growth and capacity constraints are risking an inflation pickup and decreasing short rates when the economy is in a recession. The market bond appears to incorporate expectations of such countercyclic monetary policy into the term structure. Thus, the market's expectations about short-term rate changes may be reasonably flat when the monetary policy is inactive and declining (rising) in periods when the central bank has begun to actively ease (tighten) monetary policy. Central banks often adjust short rates gradually, in trends; thus, the first easing (tightening) moves may create market expectations of further rate declines (increases). However, expectations also may have a mean-reverting component. If the market perceives certain rate levels to be “normal,” it may expect rate changes even if the monetary rate changes policy is inactive but short rates are exceptionally high (1981-82) or low (1992-93). (ii) The market's distant rate expectations probably reflect mainly its long-term inflation views, which are influenced by factors such as the country's inflation history, the size of the government debt and budget deficits (government's incentive to “monetize” debt), the strength of anti-inflationary forces (central bank independence, discipline from financial markets, political clout of long-term savers versus debt holders), and the exchange rate policy. The market's expectations about the real rates in the long run probably move quite slowly, based on its perception of the future investment-saving balance. *This report assumes that market expectations are already given, and this note briefly discusses the economic determinants of these expectations. The classic Wall Street adage—central banks determine the level of short-term yields, while market inflation expectations drive long-term yields—is too simplistic, but it still captures the main determinants of yield behavior. (1) Market participants may have more specific expectations of the near-term behavior of short-term yields than of more distant events. These expectations are closely related to market views on economic conditions and the direction of monetary policy. If the market expects economic growth and credit demand to increase, it often expects real yields and inflation to rise, and vice versa. In addition, central banks often try to influence inflation and economic growth by raising (nominal and real) short-term yields when economic growth is rapid and productivity constraints lead to a rapid rise in inflation, or by lowering short-term yields during economic downturns. The bond market seems to incorporate these counter-cyclical monetary policy expectations into the term structure. Therefore, when central bank monetary policy is not aggressive, the market expects short-term yields to change little, and when central banks begin to actively loosen (tighten) monetary policy, the market expects short-term yields to fall (rise). Central banks tend to adjust short-term yields gradually, so the first easing (tightening) measure may lead to a further decline (rise) in market expectations of yields. However, expectations may also have a mean-reverting component. If the market believes there is a certain "normal" level of yield, it may still expect yields to change when short-term yields are abnormally high (e.g., 1981-82) or low (e.g., 1992-93), even if monetary policy is not proactive at this time. (2) Market expectations of long-term yields may mainly reflect views on long-term inflation, which are influenced by factors such as the country's inflation history, the size of government debt and budget deficit (government encourages "monetization" of debt), anti-inflationary forces (central bank independence, financial market discipline, and the political influence of long-term depositors on creditors) and exchange rate policy. Based on views on the future investment-saving balance, market expectations of long-term real yields may change quite slowly*

8. It is sometimes asserted that the PEH (Predicted Returns and Hidden Returns) must hold because the existence of any near-term expected return differentials across bonds would imply arbitrage opportunities. This statement is only true in a theoretical risk-neutral world (which is often used in derivatives pricing); it is not true in the real world. Even though arbitrage arguments determine the levels of forward rates—these must *be consistent with spot rates according to Equation (3)—these arguments do not say whether forward rates reflect rate expectations or required bond risk premia. Asness presents a lucid discussion on this complex issue in “OAS Models, Expected Returns, and a Steep Yield Curve” in the Journal of Portfolio Management* , Summer 1993 . *Even if arbitrage determines the level of forward yields (according to equation (3), consistent with spot yields), these arguments do not suggest that forward yields reflect yield expectations or the required bond risk premium. Asness provides a clear discussion of this complex issue*

in "OAS Models, Expected Returns, and a Steep Yield Curve" (Journal of Portfolio Management, Summer 1993) .

9. Because short-term bonds have very little convexity, the convexity bias is only a few basis points at maturities of less than three years. Therefore, it is reasonable to ignore the bias as an approximation when analyzing only short-term bonds. For longer-term bonds, the rolling yields are clearly downward-biased estimates of expected returns because they ignore the significant positive impact of convexity . *This impact is probably the main explanation for the typically concave (humped) shape of the long end of the spot yield curve. We discuss these topics in more detail in a forthcoming report, *Convexity Bias and the Yield Curve*. We will discuss these topics in more detail in our future report, "Convexity Bias and Yield Curves.* "
10. The finding that forwards predict the "wrong" sign for long-term rate changes is not new: it was noted in many academic studies in the 1980s (see "Literature Guide"). Moreover, Frederick Macaulay, who pioneered the concept of duration remarked on this pattern already in 1938. We must caution, however, that the estimated negative relation is not very strong for the long-term bonds, the negative correlation coefficients (the first line in Figure 7) are only about one standard deviation away from zero. The estimated correlation coefficients between FSP_n and realized BRP_n (The second line in Figure 7) are more significant, about two standard deviations above zero. *The discovery that forward yields predict "incorrect" directions of long-term yield changes is not new: it was noted in many academic studies in the 1980s (see the "Literature Guide"). Furthermore, Frederick Macaulay (who introduced the concept of duration) commented on this pattern as early as 1938. However, we must note that for long-term bonds, the estimated negative correlation is not very significant; the correlation coefficient (first row in Figure 7) is only one standard deviation away from zero.* FSP_n *and implementation* BRP_n *The estimated correlation coefficient between them (second row in Figure 7) is more significant, approximately two standard deviations away from zero* .
11. Persistence factors were discussed in Global Fixed-Immune *Investments : The Persistence Effect, Martin Leibowitz, Lawrence Bader and Stanley Kogelman, Salomon Brothers Inc, February* 1994 .
12. There are two main criticisms regarding the use of survey data. First, one can argue that analyst forecasts are not representative of true market expectations. Second, we do not know exactly when the rate forecasts were made; therefore, the expected rate change is measured with error if we subtract an incorrect base (beginning) yield from the forecast. We use mid-month rates as the base yields because, according to the Wall Street *Journal, most survey responses are returned around the 15th of each month. It is not clear whether criticism should bias the forecasts systematically in one way or the other and, thus, have any significant impact on the average numbers over time.* What *remains unclear is which type of criticism caused the systematic bias in the forecasts and significantly affected the mean over a period of time.*
13. The computation is based on Equation (13) in the Appendix, adjusted for the six-month holding period. The forwards implies, on average, 106 basis points higher yield increases than the surveys. Ignoring convexity, we can approximate the impact of a 106-basis-point yield increase on bond returns by multiplying 106 with the duration bonds at horizon (for a three-month bill, "n-1" = 0.25). ($BRP_n \approx (\Delta f_{0.25} - \Delta s_{0.25}) * 0.25 = (80 - (-26)) * 0.25 = 26.5$ (basis points.) This 26.5-basis-point average risk premium is the average difference between the expected six-month holding-period return of a nine-month bill and a riskless six-month bill. After annualization, we have an estimated 53-basis-point annual risk premium for a strategy of rolling over nine-month bills every six months. *The calculation is based on equation (13) in the appendix, adjusted for a six-month holding period. On average, the implied result for forward yields is 106 basis points higher than what was found in the survey. Ignoring convexity, we can estimate the impact of the 106-basis-point yield increase by multiplying 106 by the end-of-period bond duration (for three-month Treasury bills, "n-1" = 0.25).* $BRP_n \approx (\Delta f_{0.25} - \Delta s_{0.25}) * 0.25 = (80 - (-26)) * 0.25 = 26.5$ (Basis points) This average 26.5 percentage point risk premium represents the average difference in expected returns between holding six-month and nine - *month Treasury bills for six months. Annualized, we estimate an annualized risk premium of 53 basis points for a strategy holding nine-month Treasury bills for six months.*
14. We further discuss the empirical behavior of the bond risk premium in forthcoming reports, " *Does Duration Extension Enhance Long-Term Expected Returns?* " and "Forecasting *US* Bond Returns . *"*
15. Moreover, both measures are closely related to other measures of yield curve steepness. Equations (14) and (16) show that the forward-spot premium can he written as the sum of the term spread (or the yield difference between a long rate and a short rate. $s_n - s_1$ and the rolldown return $(n-1) * (s_n - s_1)$. If the spot curve is linear, the forward-spot premium will be exactly twice the term spread, Even if the curve is not linear, the forward-spot premium and the term spread are very highly correlated. Finally, many of the

relations developed in this Appendix for zeros also hold approximately for coupon bonds if we substitute their durations for maturities (n), their end-of-horizon durations for n-1, and their yields for spot rates (s_nFurthermore , *these two measures are closely related to other measures of the steepness of the yield curve. Equations (14) and (16) show that the forward-spot premium can be written as the term spread (or the difference between long-term and short-term yields).*$s_n - s_1$*) and declining returns (*$(n-1)*(s_n - s_1)$ *The sum of (n) and (n-1) is used. If the spot yield curve is linear, the forward-spot premium will be twice the term spread, and even if the curve is not linear, the forward-spot premium and term spread are still highly correlated. Finally, if we replace term (n) with duration, n-1 with end-of-term duration, and the spot yield (n-1) with the term spread, the forward-spot premium will be twice the term spread.*s_n *If we replace) with yield to maturity, then many of the relationships derived in the appendix for zero-coupon bonds will also approximately hold true for zero-coupon bonds.* ↩

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